

Artificial Intelligence

MODULE-1: Introduction & Intelligent Agents



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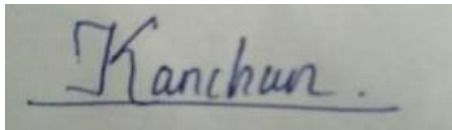
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Signature

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Artificial Intelligence

Unit I

1.1 Introduction

- 1.1.1 What is Artificial Intelligence?
- 1.1.2 Foundations of AI
- 1.1.3 History
- 1.1.4 The state of art AI today

1.2 Intelligent Agents

- 1.2.1 Agents and environment
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- 1.2.4 The structure of agents.

Q.1 What is AI?

The study of how to make computers do things at which, at the moment, people are better. Some

definitions of AI divided into four categories:

Thinking Humanly “The exciting new effort to make computers think . . . machines with minds, in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that performs functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents.” (Poole et al., 1998) “AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

What are the Categories of AI?

1) Acting humanly: *The Turing Test approach*

The computer would need to possess the following capabilities:

- **natural language processing** to enable it to communicate successfully in English;
- **knowledge representation** to store what it knows or hears;
- automated reasoning to use the stored information to answer questions and to draw new conclusions;
- **machine learning** to adapt to new circumstances and to detect and extrapolate patterns.
- **computer vision** to perceive objects, and
- **robotics** to manipulate objects and move about.

2) Thinking humanly: *The cognitive modeling approach*

Here, we must have some way of determining how humans think. We need to get inside the actual workings of human minds.

There are three ways to do this: through introspection—trying to catch our own thoughts as they go by; through psychological experiments—observing a person in action; and through brain imaging—observing the brain in action. Once we have a sufficiently precise theory of the mind, it becomes possible to express the theory as a computer program.

If the program’s input–output behavior matches corresponding human behavior, that is evidence that some of the program’s mechanisms could also be operating in humans.

3) Thinking rationally: *The “laws of thought” approach*

“Socrates is a man; all men are mortal; therefore, Socrates is mortal.” These laws of thought were supposed to govern the operation of the mind; their study initiated the field called logic.

There are two main obstacles to this approach.

First, it is not easy to take informal knowledge and state it in the formal terms required by logical notation, particularly when the knowledge is less than 100% certain.

Second, there is a big difference between solving a problem "in principle" and solving it in practice.

4) Acting rationally: *The rational agent approach*

An agent is just something that acts. A rational agent is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.

The rational-agent approach has two advantages over the other approaches.

First, it is more general than the "laws of thought" approach because correct inference is just one of several possible mechanisms for achieving rationality.

Second, it is more amenable to scientific development than are approaches based on human behavior or human thought.

Q.2 What is Turing Test?

Alan Turing, in a 1951 paper, proposed a test called "The Imitation Game" that might finally settle the issue of machine intelligence. The first version of the game he explained involved no computer intelligence whatsoever. Imagine three rooms, each connected via computer screen and keyboard to the others. In one room sits a man, in the second a woman, and in the third sits a person - call him or her the "judge". The judge's job is to decide which of the two people talking to him through the computer is the man. The man will attempt to help the judge, offering whatever evidence he can (the computer terminals are used so that physical clues cannot be used) to prove his man-hood. The woman's job is to trick the judge, so she will attempt to deceive him, and counteract her opponent's claims, in hopes that the judge will erroneously identify her as the male.

What does any of this have to do with machine intelligence? Turing then proposed a modification of the game, in which instead of a man and a woman as contestants, there was a human, of either gender, and a computer at the other terminal. Now the judge's job is to decide which of the contestants is human, and which the machine. Turing proposed that if, under these conditions, a judge were less than 50% accurate, that is, if a judge is as likely to pick either human or computer, then the computer must be a passable simulation of a human being and hence, intelligent. The game has been recently modified so that there is only one contestant, and the judge's job is not to choose between two contestants, but simply to decide whether the single contestant is human or machine.

Q.3 What kind of capabilities computer would need to possess to pass Turing test?

- natural language processing to enable it to communicate successfully in English;
- knowledge representation to store what it knows or hears;
- automated reasoning to use the stored information to answer questions and to draw new conclusions;
- machine learning to adapt to new circumstances and to detect and extrapolate patterns.

Q.4 What is required to pass total Turing test?

To pass the total Turing Test, the computer will need:

- computer vision to perceive objects, and
- robotics to manipulate objects and move about.

<https://www.youtube.com/watch?v=mJeNghZXtMo&t=202s> (What is Artificial Intelligence?)

Q.5 Define agent

An agent is just something that acts, operate autonomously, perceive their environment, persist over a prolonged time period, and adapt to change, and create and pursue goals.

Q.6 Define rational agent.

A rational agent is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.

Q.7 List the disciplines that contributed ideas, viewpoints, and techniques to AI.

Philosophy:

- Can formal rules be used to draw valid conclusions?
- How does the mind arise from a physical brain?
- Where does knowledge come from?
- How does knowledge lead to action?

Logic, methods of reasoning, mind as physical system foundations of learning, language, rationality are the foundations of philosophy. Thus, using this field, AI will be having the idea of where exactly the knowledge comes from and how does it leads to an action.

Mathematics: Formal representation and proof algorithms, computation, (un)decidability, (in)tractability, probability

- What are the formal rules to draw valid conclusions?
- What can be computed?
- How do we reason with uncertain information?

Mathematics provided the tasks to manipulate statements of logical certainty as well as uncertain probabilistic statements. It also sets the groundwork for understanding computation and reasoning about algorithm.

Economics: utility, decision theory

- How should we make decisions so as to maximize payoff?
- How should we do this when others may not go along?
- How should we do this when the payoff may be far in the future?

Economics, formalizes the problem of making decision that maximize the expected outcome to the decisionmaker.

Neuroscience: physical substrate for mental activity

- How do brains process information?

Neuroscience is the study of the nervous system, particularly the brain.

It discovered some facts about how the brain works and the ways in which it is similar to and different from the computer.

Psychology: phenomena of perception and motor control, experimental techniques

- How do humans and animals think and act?

Psychologists adopted the idea that humans and animals can be considered as information processing machines as well as it will search how human and animals think and act.

Computer engineering: building fast computers

- How can we build an efficient computer?

Computer Engineers provided the powerful machines that make AI application possible.

For artificial intelligence to succeed, we need two things: intelligence and an artifact. The computer has been the artifact of choice.

Control theory and cybernetics: design systems that maximize an objective function over time

- How can artifacts operate under their own control?

Linguistics: knowledge representation, grammar

- How does language relate to thought?

Q.8 Write a Short Note on history of AI.

The gestation of artificial intelligence (1943–1955)

The first work that is now generally recognized as AI was done by Warren McCulloch and Walter Pitts (1943). They proposed a model of artificial neurons in which each neuron is characterized as being “on” nor “off,” with a switch to “on” occurring in response to stimulation by a sufficient number of neighbouring neurons.

Donald Hebb (1949) demonstrated a simple updating rule for modifying the connection strengths between neurons. His rule, now called Hebbian learning.

The birth of artificial intelligence (1956) Princeton was home to another influential figure in AI, John McCarthy. McCarthy convinced Minsky, Claude Shannon, and Nathaniel Rochester to help him bring together U.S. researchers interested in automata theory, neural nets, and the study of intelligence.

Two researchers from Carnegie Tech, Allen Newell and Herbert Simon, had ideas and in some cases programs for particular applications such as checkers.

Newell and Simon already had a reasoning program, the Logic Theorist (LT), about which Simon claimed, “We have invented a computer program capable of thinking non-numerically, and thereby solved the venerable mind–body problem.

Early enthusiasm, great expectations (1952–1969) Newell and Simon’s early success was followed up with the General Problem Solver, or GPS.

Thus, GPS was probably the first program to embody the “thinking humanly” approach. The success of GPS and subsequent programs as models of cognition led Newell and Simon (1976) to formulate the famous physical symbol system hypothesis, which states that “a physical symbol system has the necessary and sufficient means for general intelligent action.”

A dose of reality (1966–1973)

In 1966, a report by an advisory committee found that “there has been no machine translation of general scientific text, and none is in immediate prospect.” All U.S. government funding for academic translation projects was canceled. Today, machine translation is an imperfect but widely used tool for technical, commercial, government, and Internet documents.

The illusion of unlimited computational power was not confined to problem-solving programs. Early experiments in machine evolution (now called genetic algorithms) (Friedberg, 1958; Friedberg et al., 1959) were based on the undoubtedly correct belief that by making an appropriate series of small mutations to a machine-code program, one can generate a program with good performance for any particular task.

Knowledge-based systems: The key to power? (1969–1979)

The DENDRAL researchers consulted analytical chemists and found that they worked by looking for well-known patterns of peaks in the spectrum. Example: if there are two peaks at x_1 and x_2 such that (a) $x_1 + x_2 = M + 28$ (M is the mass of the whole molecule); (b) $x_1 - 28$ is a high peak; (c) $x_2 - 28$ is a high peak; (d) At least one of x_1 and x_2 is high.

There is a ketone subgroup AI becomes an industry (1980–present)

In 1981, the Japanese announced the “Fifth Generation” project, a 10-year plan to build intelligent computers running Prolog. Overall, the AI industry boomed from a few million dollars in 1980 to billions of dollars in 1988, including hundreds of companies building expert systems, vision systems, robots, and software and hardware specialized for these purposes.

AI adopts the scientific method (1987–present)

In terms of methodology, AI has finally come firmly under the scientific method. To be accepted, hypotheses must be subjected to rigorous empirical experiments, and the results must be analysed statistically for their importance (Cohen, 1995)

The emergence of intelligent agents (1995–present)

A related idea is the subfield of Artificial General Intelligence or AGI (Goertzel and ARTIFICIAL GENERAL INTELLIGENCE Pennachin, 2007), which held its first conference and organized the Journal of Artificial General Intelligence in 2008. AGI looks for a universal algorithm for learning and acting in any environment, and has its roots in the work of Ray Solomonoff (1964), one of the attendees of the original 1956 Dartmouth conference. Guaranteeing that what we create is really FRIENDLY AI Friendly AI is also a concern (Yudkowsky, 2008; Omohundro, 2008).

The availability of very large data sets (2001–present)

“Knowledge bottleneck” in AI—the problem of how to express all the knowledge that a system needs—may be solved in many applications by learning methods rather than hand-coded knowledge engineering, provided the learning algorithms have enough data to go on (Halevy et al., 2009). Reporters have noticed the surge of new applications and have written that “AI Winter” may be yielding to a new Spring (Havenstein, 2005). As Kurzweil (2005) writes, “today, many thousands of AI applications are deeply embedded in the infrastructure of every industry.”

Q.9 Write a short note on State of Art.

OR

What are the application areas of Artificial Intelligence.

There are so many activities in so many subfields. Here we sample a few applications.

Robotic vehicles: A driverless robotic car named STANLEY sped through the rough terrain of the Mojave desert at 22 mph, finishing the 132-mile course first to win the 2005 DARPA Grand Challenge. STANLEY is a Volkswagen Touareg outfitted with cameras, radar, and laser rangefinders to sense the environment and onboard software to command the steering, braking, and acceleration. The following year CMU’s BOSS won the Urban Challenge, safely driving in traffic through the streets of a closed Air Force base, obeying traffic rules and avoiding pedestrians and other vehicles.

Speech recognition: A traveler calling United Airlines to book a flight can have the entire conversation guided by an automated speech recognition and dialog management system.

Robotics: The iRobot Corporation has sold over two million Roomba robotic vacuum cleaners for home use. The company also deploys the more rugged PackBot to Iraq and Afghanistan, where it is used to handle hazardous materials, clear explosives, and identify the location of snipers.

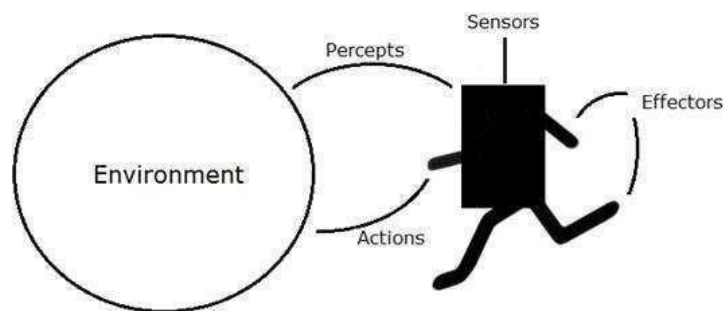
Spam fighting: Each day, learning algorithms classify over a billion messages as spam, saving the recipient from having to waste time deleting what, for many users, could comprise 80% or 90% of all messages, if not classified away by algorithms. Because the spammers are continually updating their tactics, it is difficult for a static programmed approach to keep up, and learning algorithms work best.

Game playing: IBM’s DEEP BLUE became the first computer program to defeat the world champion in a chess match when it bested Garry Kasparov by a score of 3.5 to 2.5 in an exhibition match. Kasparov said that he felt a “new kind of intelligence” across the board from him. Newsweek magazine described the match as “The brain’s last stand.”

Human champions studied Kasparov’s loss and were able to draw a few matches in subsequent years, but the most recent human-computer matches have been won convincingly by the computer.

Machine Translation: A computer program automatically translates from Arabic to English, allowing an English speaker to see the headline “Ardogan Confirms That Turkey Would Not Accept Any Pressure, Urging Them to Recognize Cyprus.” The program uses a statistical model built from examples of Arabic-to-English translations and from examples of English text totaling two trillion words (Brants et al., 2007). None of the computer scientists on the team speak Arabic, but they do understand statistics and machine learning algorithms.

Q.10 Explain the concept of agent and environment



An agent is anything that can be viewed as:

- perceiving its environment through sensors and
- acting upon that environment through actuators

Every agent can perceive its own actions (but not always the effects)

A software agent receives keystrokes, file contents, and network packets as sensory inputs and acts on the environment by displaying on the screen, writing files, and sending network packets.

Q.11 Terminology of Agent

Agent Terminology

- Performance Measure of Agent – It is the criteria, which determines how successful an agent is.
- Behavior of Agent – It is the action that agent performs after any given sequence of percepts.
- Percept – It is agent's perceptual inputs at a given instance.
- Percept Sequence – It is the history of all that an agent has perceived till date.
- Agent Function – It is a map from the precept sequence to an action.

Q.12 What is percept?

The term percept to refer to the agent's perceptual inputs at any given instant

Q.13 Define percept sequence

An agent's percept sequence is the complete history of everything the agent has ever perceived.

Q.14 Explain the rational approach of AI

Ans: Rationality:

- Rationality is nothing but status of being reasonable, sensible, and having good sense of judgment.
- Rationality is concerned with expected actions and results depending upon what the agent has perceived. Performing actions with the aim of obtaining useful information is an important part of rationality.

Rationality of an agent depends on the following four factors –

- The performance measures, which determine the degree of success.
- Agent's Percept Sequence till now.
- The agent's prior knowledge about the environment.
- The actions that the agent can carry out.

The problem the agent solves is characterized by Performance Measure, Environment, Actuators, and Sensors (PEAS).

Q.15 What is task environment?

The performance measure, the environment, and the agent's actuators and sensors (PEAS) are grouped under the heading of the task environment.

Q.16 Give the PEAS description of taxi's task environment

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, Other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Performance measure: Desirable qualities include getting to the correct destination; minimizing fuel consumption and wear and tear; minimizing the trip time or cost; minimizing violations of traffic laws and disturbances to other drivers; maximizing safety and passenger comfort; maximizing profits.

Driving environment: Any taxi driver must deal with a variety of roads, ranging from rural lanes and urban alleys to 12-lane freeways. The roads contain other traffic, pedestrians, stray animals, road works, police cars, puddles, and potholes. The taxi must also interact with potential and actual passengers. There are also some optional choices.

Actuators: include those available to a human driver: control over the engine through the accelerator and control over steering and braking. In addition, it will need output to a display screen or voice synthesizer to talk back to the passengers, and perhaps some way to communicate with other vehicles, politely or otherwise. **Sensors:** include one or more controllable video cameras so that it can see the road; it might augment these with infrared or sonar sensors to detect distances to other cars and obstacles. To avoid speeding tickets, the taxi should have a speedometer, and to control the vehicle properly, especially on curves, it should have an accelerometer. To determine the mechanical state of the vehicle, it will need the usual array of engine, fuel, and electrical system sensors. Like many human drivers, it might want a global positioning system (GPS) so that it doesn't get lost. Finally, it will need a keyboard or microphone for the passenger to request a destination.

<https://www.youtube.com/watch?v=gEy91PGGLR0> (Autonomous car / self-driving car - How it works!)

Q.17 Various agents and their performance descriptions

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Q.18 What are the different Properties of task environments?

a) **Fully observable vs. partially observable:**

- If an agent's sensors give it access to the complete state of the environment at each point in time, then we say that the task environment is fully observable. A task environment is effectively fully observable if the sensors detect all aspects that are relevant to the choice of action; relevance, in turn, depends on the performance measure. Fully observable environments are convenient because the agent need not maintain any internal state to keep track of the world.
- An environment might be partially observable because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data—for example, a vacuum agent with only a local dirt sensor cannot tell whether there is dirt in other squares, and an automated taxi cannot see what other drivers are thinking. If the agent has no sensors at all then the environment is unobservable.

b) **Single agent vs. multiagent:**

- The distinction between single-agent and multiagent environments may seem simple enough.

c) **Deterministic vs stochastic:**

- If the next state of the environment is completely determined by the current state and the action executed by the agent, then we say the environment is deterministic; otherwise, it is stochastic.
- d) **Episodic vs. sequential:**
- In an episodic task environment, the agent's experience is divided into atomic episodes. In each episode the agent receives a percept and then performs a single action. Crucially, the next episode does not depend on the actions taken in previous episodes. Many classification tasks are episodic.
 - In sequential environments, on the other hand, the current decision could affect all future decisions. Chess and taxi driving are sequential: in both cases, short-term actions can have long-term consequences. Episodic environments are much simpler than sequential environments because the agent does not need to think ahead.
- e) **Static vs. dynamic:**
- If the environment can change while an agent is deliberating, then we say the environment is dynamic for that agent; otherwise, it is static. Static environments are easy to deal with because the agent need not keep looking at the world while it is deciding on an action, nor need it worry about the passage of time.
 - Dynamic environments, on the other hand, are continuously asking the agent what it wants to do; if it hasn't decided yet, that counts as deciding to do nothing
- f) **Discrete vs. continuous:**
- The discrete/continuous distinction applies to the state of the environment, to the way time is handled, and to the percepts and actions of the agent. For example, the chess environment has a finite number of distinct states
- g) **Known vs. unknown:**
- In a known environment, the outcomes (or outcome probabilities if the environment is stochastic) for all actions are given.
 - Obviously, if the environment is unknown, the agent will have to learn how it works in order to make good decisions.

Q.19 Give some examples of task environments with their characteristics?

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi Driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical Diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image Analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-Picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery Controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English Controller	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Q.20 Explain the structure of Agents?

1. The job of AI is to design an agent program that implements the agent function—the mapping from percepts to actions. We assume this program will run on some sort of computing device with physical sensors and actuators—we call this the architecture:

$$\text{agent} = \text{architecture} + \text{program}$$
2. **Agent programs:**
 The agent programs that we design in this book all have the same skeleton: they take the current percept as input from the sensors and return an action to the actuators. Notice the difference between the agent

program, which takes the current percept as input, and the agent function, which takes the entire percept history. The agent program takes just the current percept as input because nothing more is available from the environment; if the agent's actions need to depend on the entire percept sequence, the agent will have to remember the percepts.

```

function TABLE-DRIVEN-AGENT (percept) returns an action
persistent: percepts, a sequence, initially empty
               table, a table of actions, indexed by percept sequences, initially fully
               specified
append percept to the end of percepts
action ← LOOKUP (percepts, table)
return action

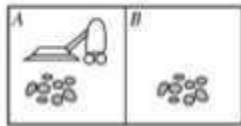
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The TABLE-DRIVEN-AGENT Program is invoked for each new percept and returns an action each time. It retains the complete percept sequence in memory.

Q.21 Define agent and environment with respect to vacuum cleaner world

This world is so simple that we can describe everything that happens; it's also a made-up world, so we can invent many variations. This particular world has just two locations: squares A and B. The vacuum agent perceives which square it is in and whether there is dirt in the square. It can choose to move left, move right, suck up the dirt, or do nothing. One very simple agent function is the following: if the current square is dirty, then suck; otherwise, move to the other square. A partial tabulation of this agent function.

Vacuum-cleaner world



- Percepts: location and contents e.g. [A, Dirty]
- Actions: Left, Right, Suck
- Function-table (table look-up agent)

Percept	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck

Q.22 Explain different types of agent programs.

The job of AI is to design an agent program that implements the agent function—the mapping from percepts to actions. We assume this program will run on some sort of computing device with physical sensors and actuators—we call this the architecture :

$$\text{agent} = \text{architecture} + \text{program} .$$

Four basic kinds of agent programs that embody the principles underlying almost all intelligent systems:

- Simple reflex agents;
- Model-based reflex agents;
- Goal-based agents;
- Utility-based agents.

Each kind of agent program combines particular components in particular ways to generate actions.

1. Simple reflex agents: The simplest kind of agent is the simple reflex agent. These agents select actions on the basis of the current percept, ignoring the rest of the percept history.

Simple reflex agents have the admirable property of being simple, but they turn out to be of limited intelligence. The agent function is based on condition-action rule.

This agent function is only successful if the environment is fully observable.

2. Model-based reflex agents: It works by finding a rule whose condition matches the current situation. A model-based agent can handle partially observable environments by use of model about the world. The agent has to keep track of internal state which is adjusted by each percept and that depends on the percept history. The current state is stored inside the agent which maintains some kind of structure describing the part of the world which cannot be seen. Updating the state requires the information about:

- how the world evolves in-dependently from the agent, and
- how the agent actions affects the world.

3. Goal-based agents: They choose their actions in order to achieve goals.

Goal-based approach is more flexible than reflex agent since the knowledge supporting a decision is explicitly modelled, thereby allowing for modifications.

Goal – It is the description of desirable situations.

The goal-based agent's behaviour can easily be changed to go to a different destination, simply by specifying that destination as the goal.

4. Utility-based agents: The agents which are developed having their end uses as building blocks are called utility based agents. They choose actions based on a preference (utility) for each state.

Sometimes achieving the desired goal is not enough. Agent happiness should be taken into consideration. Utility describes how “happy” the agent is. Because of the uncertainty in the world, a utility agent chooses the action that maximizes the expected utility. A utility function maps a state onto a real number which describes the associated degree of happiness.

Q.23 Give the agent program for a simple reflex agent in the two-state vacuum environment. function REFLEX-VACUUM-AGENT([location,status]) returns an action

```

if status = Dirty then return Suck else if
location = A then return Right else if
location = B then return Left

```

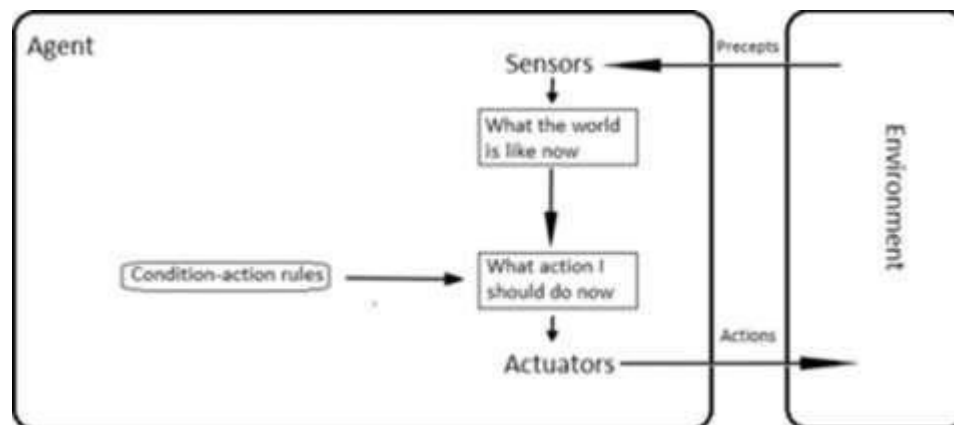
Q.24 Explain Simple reflex agent.

The simplest kind of agent is the simple reflex agent. These agents select actions on the basis of the current percept, ignoring the rest of the percept history.

Simple reflex agents have the admirable property of being simple, but they turn out to be of limited intelligence.

The agent function is based on condition-action rule.

This agent function is only successful if the environment is fully observable.



Problems with simple reflex agents:-

- Very limited intelligence.
- No knowledge of non-perceptual parts of state.
- Usually too big to generate and store.
- If there occurs any change in the environment, then the collection of rules need to be updated.

Q.25 Write the function for simple reflex agent.

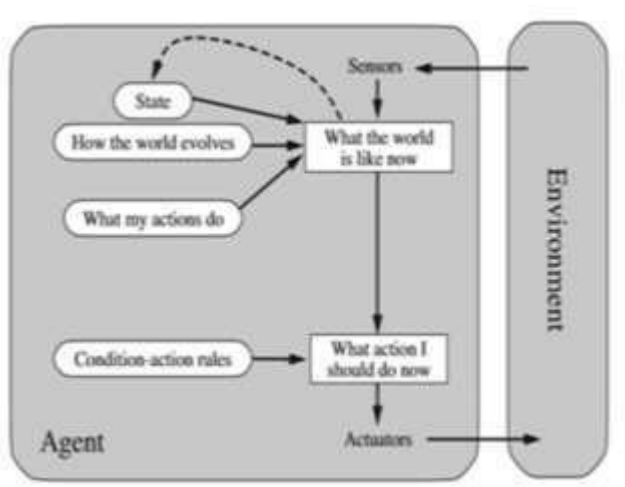
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function SIMPLE-REFLEX-AGENT(percept ) returns an action
  persistent: rules, a set of condition–action rules
  state ← INTERPRET-INPUT(percept )
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
return action

```

Q.26 Explain model based reflex agents.

It works by finding a rule whose condition matches the current situation. A model-based agent can handle partially observable environments by use of model about the world. The agent has to keep track of internal state which is adjusted by each percept and that depends on the percept history. The current state is stored inside the agent which maintains some kind of structure describing the part of the world which cannot be seen.



Updating the state requires the information about:

- how the world evolves in-dependently from the agent, and
- how the agent actions affects the world

Q.27 Write a function for model based reflex agent

```

function MODEL-BASED-REFLEX-AGENT(percept ) returns an action
  persistent: state, the agent's current conception of the world state
  model, a description of how the next state depends on current state and actionrules, a set of condition–
  action rules
  action, the most recent action, initially none
  state ← UPDATE-STATE(state, action, percept ,model )
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
return action

```

Q.28 Explain Goal based agents.

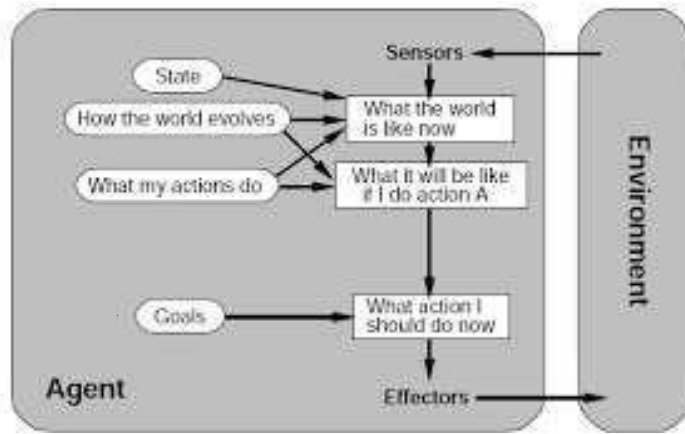
They choose their actions in order to achieve goals.

Goal-based approach is more flexible than reflex agent since the knowledge supporting a decision is explicitly modelled, thereby allowing for modifications.

Goal – It is the description of desirable situations.

The goal-based agent's behaviour can easily be changed to go to a different destination, simply by specifying that destination as the goal.

Sometimes goal-based action selection is straightforward—for example, when goal satisfaction results immediately from a single action. Sometimes it will be more tricky—for example, when the agent has to consider long sequences of twists and turns in order to find a way to achieve the goal.



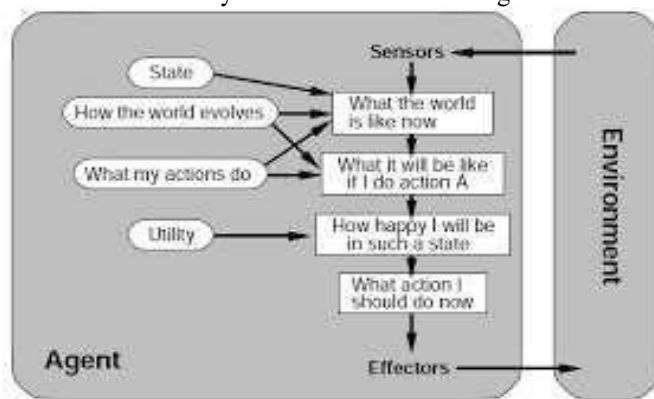
Q.29 Explain utility based agents.

The agents which are developed having their end uses as building blocks are called utility based agents. They choose actions based on a preference (utility) for each state.

Sometimes achieving the desired goal is not enough. Agent happiness should be taken into consideration. Utility describes how "happy" the agent is. Because of the uncertainty in the world, a utility agent chooses the action that maximizes the expected utility.

A utility function maps a state onto a real number which describes the associated degree of happiness.

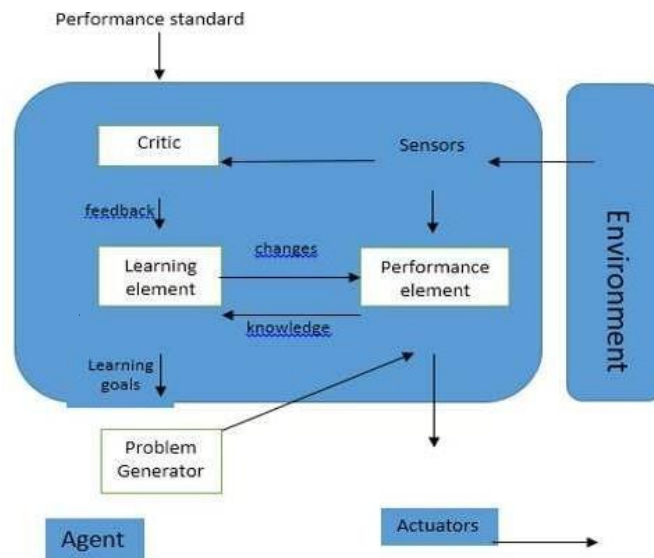
If the internal utility function and the external performance measure are in agreement, then an agent that chooses actions to maximize its utility will be rational according to the external performance measure.



Q.30 Explain learning agent?

Ans. A learning agent can be divided into four conceptual components which are the basic building blocks of Learning Agent.

1. Performance element
2. Learning element
3. Critic
4. Problem generator



1. Performance element

- Responsible for choosing actions that are known to offer good outcomes
- Corresponds to the agents discussed earlier

2. Learning element

- Responsible for improving the performance element
- Requires feedback on how well the agent is doing

3. Critic

- Responsible for providing feedback
- Compares outcomes with some objective performance standard from outside the agent

4. Problem generator

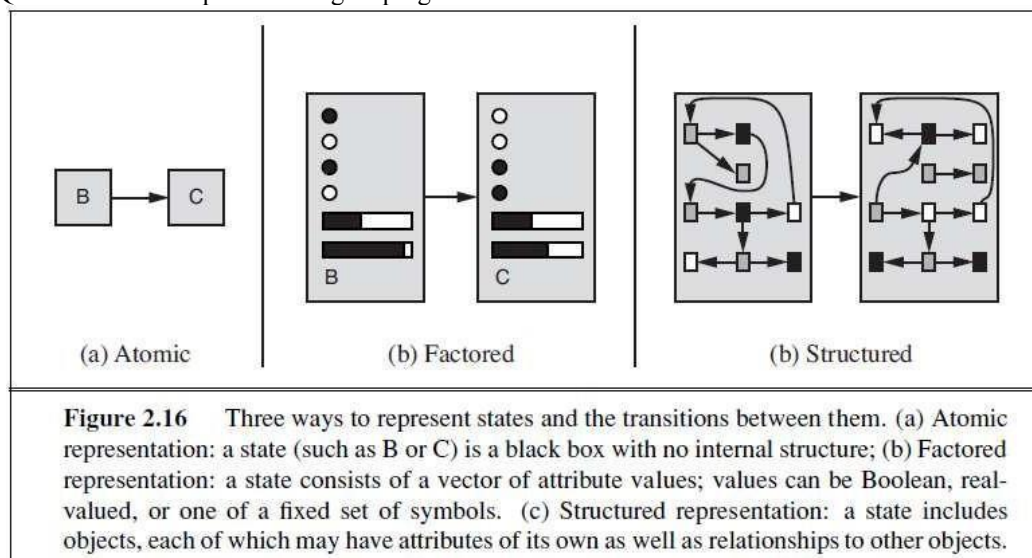
- Responsible for generating new experience
- Requires exploration –trying unknown actions which may be sub-optimal

Example:

Consider a taxi driver agent

- **Performance element:** you want to go into Mumbai CST? Let's take Highway, it's worked well previously
- **Problem generator:** nah, let's try Freeway for a change –it may be better
- **Critic element:** great, it was 15 minutes quicker, and what a nice view!
- **Learning element:** yeah, in future we'll take Freeway

Q.31 How the components of agent program works?



The components of agent program works in three ways i.e. atomic, factored, structured.

1. **In an atomic representation**, each state of the world is indivisible-it has no internal structure. Consider the problem of finding a driving route from one end of a country to the other via some sequence of cities. For the purposes of solving this problem, it may suffice to reduce the state of world to just the name of city we are in-asingle atom of knowledge; the algorithm underlying search and game playing, Hidden Markov models and Markov decision processes all work with atomic representations or they treat representations as if they were atomic.

2. **A factored representation** splits up each state into a fixed set of variables or attributes, each of which can have a value. Many important areas of AI are based on factored representations, including constraint satisfaction algorithms, propositional logic, planning, Bayesian networks and the machine learning algorithms.

3. **A structured representation**, in which objects and their various and varying relationships can be described explicitly. Structured representations underlie relational databases and first order logic, first order probability models, knowledge based learning and natural language understanding.

Viva Questions

1. What is AI?
2. What kind of capabilities computer would need to possess to pass Turing test?
3. What is required to pass total turing test?
4. List the disciplines that contributed ideas, viewpoints, and techniques to AI.
5. How AI developed itself? or History of AI.
6. What are the various application areas of AI?
7. Define agent
8. Define percept
9. Define percept sequence
10. Define rational agent
11. What is rationality?
12. Expand PEAS
13. Give the PEAS description for the taxi's task environment.
14. Give examples of agent types and their PEAS descriptions.
15. What are Fully observable, partially observable and unobservable task environment?
16. Single agent vs. multiagent
17. Deterministic vs. stochastic
18. Episodic vs. sequential
19. Static vs. dynamic
20. Discrete vs. continuous
21. Known vs. unknown
22. Give examples of task environments and their characteristics.
23. What are the basic kinds of agent programs
24. Define Simple reflex agents
25. Give the function of simple reflex agent
26. Define Model-based reflex agents
27. Give the function of Model-based reflex agent
28. Define Goal-based agents
29. Define Utility-based agents
30. What is utility function?
31. What are the components of learning agent?

Fill in the Blanks

1. Turing test was designed to provide a satisfactory operational definition of intelligence.
2. The interdisciplinary field of cognitive science brings together computer models from AI and experimental techniques from psychology to construct precise and testable theories of the human mind.
3. An agent is just something that acts, operate autonomously, perceive their environment, persist over a prolonged time period, adapts to change, and create and pursue goals.
4. A rational agent is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.
5. An alternative to dualism is materialism, which holds that the brain's operation according to the laws of physics constitutes the mind.

6. Decision theory, which combines probability theory with utility theory, provides a formal and complete framework for decisions (economic or otherwise) made under uncertainty— that is, in cases where probabilistic descriptions appropriately capture the decision maker's environment.
7. Neuroscience is the study of the nervous system, particularly the brain.
8. Cognitive psychology, which views the brain as an information-processing device, can be traced back at least to the works of William James
9. Craik specified the three key steps of a knowledge-based agent: (1) the stimulus must be translated into an internal representation, (2) the representation is manipulated by cognitive processes to derive new internal representations, and (3) these are in turn retranslated back into action.
10. Modern control theory, especially the branch known as stochastic optimal control, has as its goal the design of systems that maximize an objective function over time.
11. The perceptron convergence theorem says that the learning algorithm can adjust the connection strengths of a perceptron to match any input data, provided such a match exists.
12. Genetic algorithms were based on the undoubtedly correct belief that by making an appropriate series of small mutations to a machine-code program, one can generate a program with good performance for any particular task.
13. An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.
14. The term percept is used to refer to the agent's perceptual inputs at any given instant.
15. An agent's percept sequence is the complete history of everything the agent has ever perceived.
16. An agent's behavior is described by the agent function that maps any given percept sequence to an action.
17. A rational agent is one that does the right thing.
18. The notion of desirability captured by performance measure evaluates any given sequence of environment states.
19. For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.
20. An omniscient agent knows the actual outcome of its actions and can act accordingly.
21. Some software agents (or software robots or softbots) exist in rich, unlimited domains.
22. An environment might be partially observable because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data.
23. If the agent has no sensors at all then the environment is unobservable.
24. Chess is a competitive multiagent environment.
25. In the taxi-driving environment, avoiding collisions maximizes the performance measure of all agents, so it is a partially cooperative multiagent environment.
26. An environment is uncertain if it is not fully observable or not deterministic.
27. A nondeterministic environment is one in which actions are characterized by their possible outcomes, but no probabilities are attached to them.
28. In an episodic task environment, the agent's experience is divided into atomic episodes.
29. If the environment can change while an agent is deliberating, then we say the environment is dynamic for that agent.
30. If the environment itself does not change with the passage of time but the agent's performance score does, then we say the environment is semidynamic.
31. The code repository also includes an environment generator for each environment class that selects particular environments (with certain likelihoods) in which to run the agent.
32. Agent = Architecture + Program
33. Simple Reflex Agents These agents select actions on the basis of the current percept, ignoring the rest of the percept history.
34. An agent that uses a model is called a model-based agent.
35. Search and planning are the subfields of AI devoted to finding action sequences that achieve the agent's goals.
36. Because "happy" does not sound very scientific, economists and computer scientists use the term utility instead.
37. An agent's utility function is essentially an internalization of the performance measure.
38. The learning element is responsible for making improvements in the learning agent.
39. The performance element is responsible for selecting external actions in the learning agent.

40. The learning element uses feedback from the critic on how the agent is doing and determines how the performance element should be modified to do better in the future.
41. In an atomic representation, each state of the world is indivisible—it has no internal structure.
42. A factored representation splits up each state into a fixed set of variables or attributes, each of which can have a value.

Graded Questions

- Q.1 What is AI?
- Q.2 What are the categories of AI?
- Q.3 What is Turing Test? Explain
- Q.4 Explain the foundations of AI
- Q.5 Write short note on history of AI
- Q.6 Write short note on State of Art
- Q.7 What are the application areas of AI? Explain
- Q.8 Define agent and environment
- Q.9 Define agent and environment with respect to vacuum cleaner world
- Q.10 Write short note on concept of rationality.
- Q.11 How do you specify the task environment? Explain
- Q.12 Explain the term PEAS
- Q.13 Explain PEAS description of task environment for automated taxi
- Q.14 Explain the properties of task environments
- Q.15 Explain different types of task environments.
- Q.16 Give some examples of task environments with their characteristics
- Q.17 Give comparison between Full observable and partially observable agent
- Q.18 Explain the structure of Agents
- Q.19 Explain different types of agent programs
- Q.20 Explain Simple reflex agent
- Q.21 Explain model based reflex agents
- Q.22 Explain Goal based agents
- Q.23 Explain utility based agents
- Q.24 Explain learning agents
- Q.25 How the components of agent programs work?

MCQ's

1. In an episodic task environment, the agent's experience is divided into _____ episodes.
 a) **atomic** b) deterministic c) reflex d) sequential
2. If the environment can change while an agent is deliberating, then we say the environment is _____ for that agent.
 a) static **b) dynamic** c) sequential d) known
3. If the environment itself does not change with the passage of time but the agent's performance score does, then we say the environment is _____.
 a) static b) dynamic **c) semidynamic** d) deterministic
4. The code repository also includes an _____ for each environment class that selects particular environments (with certain likelihoods) in which to run the agent.
 a) **environment generator** b) reflex c) static d) robotic agent

5. Agent = Architecture +

- a) Simple b) **Program** c) Algorithm d) Database

6. _____ These agents select actions on the basis of the current percept, ignoring the rest of the percept history.

a) **Simple Reflex Agents**

b) utility

c) model reflex

d) Goal based

7. An agent that uses a _____ is called a model-based agent.

- a) Goal b) **Model** c) Utility d) Simple

8. _____ and _____ are the subfields of AI devoted to finding action sequences that achieve the agent's goals.

- a) Information, data b) Process, Program c) **Search, planning** d) Simple, complex

9. Because "happy" does not sound very scientific, economists and computer scientists use the term _____ instead.

- a) Goal b) Model c) **Utility** d) Simple

10. An agent's _____ is essentially an internalization of the performance measure.

- a) Goal b) Model c) Utility d) **utility function**

11. The _____ is responsible for making improvements in the learning agent.

- a) program b) **learning element** c) utility d) critic

12. The performance element is responsible for selecting external actions in the learning agent.

- a) program b) learning element c) utility d) **performance element**

13. The learning element uses feedback from the _____ on how the agent is doing and determines how the performance element should be modified to do better in the future.

- a) program b) learning element c) utility d) **critic**

14. In an _____ representation, each state of the world is indivisible—it has no internal structure.

- a) program b) learning element c) utility d) **atomic**

15. A _____ representation splits up each state into a fixed set of variables or attributes, each of which can have a value.

- a) **factored** b) atomic c) divisible d) lateral